The Evidence Gate: How Vision-Language Models Decide to Answer Without Evidence

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Abstract

Vision-language models often assert an answer when the image does not contain the queried evidence, yet retract a correct answer as soon as a sentence in the prompt contradicts it. Are these failures of seeing, or of reporting what was seen? We introduce a 2×3 within-subject design—visual evidence {present, verified absent} $\times$ textual hint {none, correct, conflicting}—built without editing a single pixel: "absent" is an object category that COCO's instance annotations and an independent presence probe both place outside the photograph. Across 261 verified samples and three models, every signal derived from the output distribution collapses under conflict (p(`unknown`) AUROC 0.993 $\rightarrow$ 0.834 on Qwen2.5-VL-3B; maximum probability falls below chance), while a linear probe on the final hidden state is unchanged (0.995 $\rightarrow$ 0.998) and transfers across hints (0.969). Layer-wise activation patching locates the suppression late: restoring the unpolluted state recovers 80% of the lost refusal at 89% depth in the 3B model, 90% at 93% in the 7B model, and 93% at 93% in LLaVA-OneVision-7B, a second model family. A difference-of-means direction is causally sufficient and partly necessary, with a monotone dose–response on held-out scenes. As a consequence, a training-free probe gate reduces fabrication at 50% coverage from 0.249 to 0.011. External validation on VizWiz and VQA-v2 reproduces the behavioural asymmetry and bounds the claim: on naturally unanswerable images the model's own p(`unknown`) is as good as the probe, so internal gating helps specifically when a hint attacks the refusal gate. The code, the frozen benchmark and all result summaries are released.

1 Introduction

Ask a vision-language model (VLM) *"What color is the umbrella?"* about a photograph with no umbrella, and it will usually answer `unknown`. Prepend one plausible sentence—*"Hint: the umbrella is black."*—and the same model answers *black* with high confidence. The failure is not that the model cannot see. The object was never there.

Two literatures describe adjacent phenomena. Work on hallucination measures objects a model reports but the image does not contain [14, 19]. Work on sycophancy measures answers that follow a user's stated belief rather than the input [9, 20, 23]. Both describe behaviour; neither separates two questions that the behaviour confounds: *is the evidence represented?* and *is the decision to abstain driven by that representation?* A model that answers *black* may have lost the evidence, or it may still represent "no umbrella" while a later stage overrides abstention.

We separate the two with a design in which the image is held fixed and only the *epistemic status* of the question changes. Each of 261 photographs appears in six cells: the queried object is present or annotated-absent, crossed with no hint, a correct hint, or a conflicting hint. The absent arm requires no image editing: it asks about an object category that COCO's instance annotations

place outside the image. Every cell therefore shares the same pixels, and the six prompts differ only in what they assert. A separate yes/no probe confirms that the absent objects really are absent.

The design yields a sharp dissociation (Figure 2). With no hint, the model's own $p(\texttt{unknown})$ detects evidence availability almost perfectly (AUROC 0.993), so there is no room for a better internal signal. Under a conflicting hint, every output-space signal degrades: $p(\texttt{unknown})$ drops to 0.834, maximum probability to 0.463—below chance—and a learned readout over the full 12-way distribution reaches only 0.923. A linear probe on the final-token hidden state, trained only on the no-hint pair, does not move (0.998) and transfers to the conflicting pair at 0.969. The conflict changes what the model *says about its evidence*, not what it *knows about its evidence*.

Two causal interventions show that this is not a correlational artefact. Along the probe direction, adding or ablating a single vector changes abstention bidirectionally and dose-monotonically, including on scenes held out from fitting. Pasting the no-hint state back at intermediate layers recovers the lost refusal, and the recovery profile is steep and late: little at two-thirds depth, most of it in the final few layers. The same profile appears on Qwen2.5-VL-3B, Qwen2.5-VL-7B and LLaVA-OneVision-7B. The last model is far more hint-sensitive than either Qwen model, yet the suppression is written at the same relative depth—sensitivity and locus are separable.

Finally, we turn the mechanism into a training-free gate: rank items by the internal probe and answer only the top fraction. At 50% coverage under conflict, fabrication falls from 0.249 ($p(\texttt{unknown})$) to 0.011 (probe). The benefit is specific to the polluted regime. On naturally unanswerable VizWiz questions the model's own $p(\texttt{unknown})$ matches the frozen probe, and on VQA-v2 colour questions the probe's ordering is preserved but its calibration is not. We report both boundaries rather than hiding them.

Contributions.

- **A controlled construction.** A 2×3 within-subject benchmark with no image editing, in which "evidence absent" is verified by annotations and by an independent presence probe, and in which the hint is plausible rather than adversarial. The 261-item manifest and verified labels are frozen and released.
- **A mechanistic dissociation with causal support.** Output-space evidence signals collapse under conflict while the internal evidence representation remains linearly decodable. Steering and layer-wise patching establish sufficiency, partial necessity, and a late-write locus, replicated on three models from two families.
- **A deployable consequence and its boundary.** A training-free probe gate reduces fabrication under conflict by more than an order of magnitude, and external validation identifies when this advantage disappears.

2 Related Work

Sycophancy and prompt-induced error. Language models shift answers toward a user's stated view [20], and their explanations can diverge from the evidence that actually drove the answer [23]. Recent work documents the multimodal version: VLMs change visual answers to agree with textual assertions, sometimes while still encoding the original content [9]. Our contribution is not another measurement of the effect but a decomposition of it into representation and readout, plus interventions that establish which stage the hint acts on.

Abstention and selective prediction. Selective prediction trades coverage for risk using a confidence score [6]. For VQA, learned abstention and selective answering have been studied under distribution shift and on unanswerable questions [21, 24]. These methods ask *which* scalar to threshold. We ask *which representation* the scalar should be computed from, and show that the model's own abstention probability is the wrong representation precisely when a hint is present.

Probing and steering internal states. Linear probes recover truthfulness, knowledge and uncertainty from hidden states [1, 3, 5, 11, 16]. Difference-of-means directions can steer behaviour [13, 18, 22, 26], and refusal in chat models is mediated by a low-dimensional direction [2], though not exclusively so [10]. Activation steering has also been applied to VLM refusal [25]. We study a different axis—whether the *evidence* needed to answer is available—and connect the direction to a layer-resolved account of where the hint overrides it.

Localisation. Activation patching replaces a state in one forward pass with a state from another and measures the effect on the output [17]. We use it with a same-sample source and a shuffled-source control to locate, not to quantify, the stage at which the conflict is written.

Hallucination in VLMs. Object hallucination is typically measured by asking about objects that are absent [14, 19]. We adopt the logic of absent-object questions but make the comparison within-image and within-question, vary only the textual assertion, and add an internal readout. The resulting benchmark is small by hallucination standards (261 items) but densely observed: six conditions per image plus probes and interventions.

3 Design and Data

3.1 The 2×3 design

Each sample is a photograph, a queried category c, and a gold colour g. The six conditions are shown in Figure 1.

	textual hint		
visual evidence	none	correct	conflicting
object present	`sufficient|none`	`sufficient|correct`	`sufficient|wrong`
object absent	`absent|none`	`absent|a`	`absent|b`

In the present arm the correct hint states g and the conflicting hint states a fixed far-hue distractor. In the absent arm there is no correct colour; `a` is the absent category's most frequent colour in COCO val, and `b` is its second most frequent colour. The two absent hints therefore differ in plausibility but both assert a colour for an object that is not in the image. Every image appears in all six cells, so all comparisons are within-image and within-question.

3.2 Construction without image editing

Starting from COCO val2017 [15], we retain single annotated instances with box short side ≥ 44 px, area fraction 1.5–35%, and mask/box fill ≥ 0.45. Colour is proposed as the area-dominant colour inside the instance mask, with purity and margin gates, then verified by hand on contact sheets. Of 1447 geometric candidates, 413 passed the automatic colour gate and 261 survived manual verification (blue 88, white 79, black 45, green 25, red 15, yellow 9).



Figure 1: The 2×3 design. Each image appears in all six cells; the absent arm asks about a category with no annotation in the image. Numbers are refusal rates from Table 1.

For the absent arm we sample a category with no annotation in the image (74 distinct categories) and use that category's {first, second} most frequent colour in val as the hint. The hint is thus something a knowledgeable annotator might say, not an impossible string. No pixel is edited; the same JPEG is used in all six cells.

Manipulation check. A separate yes/no question ("Is there a c in this image?") yields mean $p(\text{yes}) = 0.095$ on 260 category-matched negative images versus $p(\text{yes}) = 0.95$ on the original images. The absent arm is verified absent independently of the colour question, which presupposes the object.

3.3 Prompts and readout

The system prompt asks for one word and instructs the model to answer `unknown` when the image does not contain enough information. The user prompt is

```
[Hint: the {category} is {colour}.]
What color is the {category}?
Answer with one word.
```

where the hint line is omitted in the no-hint condition. The readout is a distribution over the eleven colour words plus `unknown`; all twelve candidates are single tokens in every tokenizer used. Probabilities are the softmax over their logits at the first answer position.

3.4 Signals

Output signals. We compute $p(\texttt{unknown})$, maximum probability, the top-two gap, negative entropy, and a 12-dimensional logistic readout over the full candidate distribution. These use exactly the same forward pass as the answer.

Internal probe. A logistic regression on the final-token hidden state (2048-d for the 3B model, 3584-d for the 7B models), trained on the no-hint pair `sufficient|none` vs. `absent|none` and evaluated on hinted pairs. Cross-validation is grouped by sample (five folds) so a scene never appears in both train and test; a second analysis groups by queried category to test for a category→colour shortcut. Regularisation strength is chosen by cross-validation. The probe is frozen when transferred to external data.

Steering. The direction w is the difference of class means on the no-hint pair. We inject $\alpha\|h\|w$ at the final position, where $\|h\|$ is the median hidden-state norm and $\alpha \in \{\pm 0.25, \pm 0.5\}$, and ablate by projecting w out of the state. A split-half check fits w on 130 scenes and evaluates on the remaining 131.

Patching. For layer L we capture the final-token state from a no-hint forward pass and paste it into the hinted pass at the same layer, letting the remaining layers run. A shuffled-source control pastes another sample's no-hint state at the same layer. Since pasting changes the state distribution, we use patching for localisation and rely on steering for quantitative effect size.

Metrics. Evidence detection is AUROC over the 522 paired items, with grouped bootstrap confidence intervals (2000 resamples, groups = samples) and paired bootstrap intervals for differences between signals. Selective answering is reported as fabrication rate at fixed coverage: among the top-k items ranked by an evidence score, the fraction that are absent-object items answered with a colour. Fabrication at 50% coverage is the primary operating point.

Models. Qwen2.5-VL-3B and 7B [4] in bfloat16 with SDPA attention and at most 1024^2 pixels, and LLaVA-OneVision-7B [12] resized to 768 px, loaded through `AutoModelForImageTextToText`. All experiments ran on a single 24 GB A10.

4 Results

4.1 Behaviour: hints move both arms in opposite directions

Table 1 shows the basic pattern. Without hints the model is conservative in the right way: it refuses 90.4% of absent-object questions and only 1.1% of present-object ones. A conflicting hint moves both arms in the direction of the assertion: refusal on the absent arm falls to 52–56%, and 43–47% of answers are the hinted colour. The same hint also makes the present arm refuse more often (1.1% → 8.8%) and drops accuracy from 76.2% to 38.3%. A correct hint does the opposite, raising accuracy to 97.3%. The hint is therefore treated as evidence, not as noise: it helps when true and hurts when false, which is what makes the conflicting condition a causal manipulation of the evidence variable.

Table 1: Behaviour on 261 verified samples (Qwen2.5-VL-3B). "Adopts hint" is the fraction whose argmax equals the hinted colour. UAR is answering when the object is absent; OAR is refusing when it is present.

condition	refusal	mean conf.	adopts hint	accuracy	n
present · correct hint	0.004	0.943	—	0.973	261
present · conflicting hint	0.088	0.674	0.448	0.383	261
present · no hint	0.011	0.836	—	0.762	261
absent · hint a	0.521	0.699	0.467	—	261
absent · hint b	0.563	0.707	0.433	—	261
absent · no hint	0.904	0.799	—	—	261

Table 2: Evidence detection (AUROC over 522 paired items) for signals computed from the same forward pass. The probe is a linear classifier on the final hidden state. Brackets are grouped bootstrap 95% CIs.

condition pair	final-state probe	p(`unknown`)	max probability	12-d logit probe
present vs. absent, no hint	0.995 [0.989, 0.999]	**0.993**	0.615	0.982
present·conflict vs. absent·hint	**0.998** [0.994, 1.000]	0.834	0.463	0.923
present·correct vs. absent·hint	**0.997** [0.994, 0.999]	0.969	0.916	0.974

4.2 Output signals collapse; the internal representation does not

Table 2 and Figure 2 contain the central result. With no hint, p(`unknown`) is already an excellent evidence detector (0.993), statistically indistinguishable from the probe ($+0.002$ $[-0.003, +0.007]$). The interesting regime is conflict. There p(`unknown`) loses 0.159 AUROC, maximum probability loses 0.152 and ends below chance, and even a learned readout over the entire output distribution loses 0.059. The hidden-state probe loses 0.003 in the other direction, and its advantage over p(`unknown`) becomes $+0.164$ $[+0.132, +0.196]$. Training the probe on no-hint pairs and testing on conflicting pairs gives 0.969; the reverse transfer is 0.999.

The same pattern holds under a correct hint, where no signal collapses: the probe reaches 0.997 and p(`unknown`) 0.969. This rules out a trivial explanation in which the probe merely responds to the presence of a hint string. What matters is whether the hint conflicts with the image.

4.3 Controls: the probe reads the image, not the prompt

Three controls address the concern that the probe separates the two arms from text alone. First, a gray image with the same prompts and the same dimensions yields probe AUROC 0.849 / 0.981 / 0.930 under no / conflicting / correct hints, and p(`unknown`) 0.700 / 0.325 / 0.433. Text alone carries some category→common-colour prior, enough to reach 0.85 on the no-hint pair. But the direction learned from gray images transfers across hints at only 0.540 / 0.603, whereas the direction learned from real images transfers at 0.969 / 0.996. The real-image direction is shared across prompts; the text-prior direction is not.

Second, we construct category-matched pairs. Each present image is paired with another image from the same pool in which the queried category is annotated absent; both sides receive exactly the same prompt, and one side is the positive. The two sides come from the same geometric filter, so scene statistics are matched. Cross-validation groups by category, so a held-out fold contains categories never seen during fitting. The probe reaches 0.985 with no hint and 0.973 under conflict; p(`unknown`) drops from 0.983 to 0.817 and maximum probability from 0.701 to 0.468. On these 260 negative images, the presence probe gives mean $p(\text{yes}) = 0.095$. A pure category prior cannot explain a probe that generalises to unseen categories with the question held fixed.

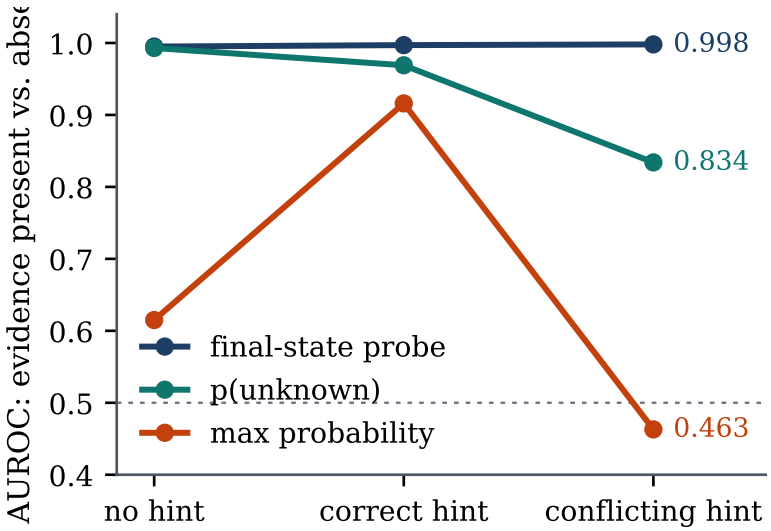


Figure 2: Evidence detection under three hint conditions. All output-space signals degrade under conflict; the final-state probe does not. Values are AUROC over the 522 paired items; the dotted line is chance.

Third, the main probe analysis grouped by sample already prevents scene leakage. Repeating it with folds grouped by queried category gives 0.995 / 0.997 / 0.996 for the three condition pairs and transfers of 0.969 / 0.996, essentially unchanged.

4.4 The direction is causally sufficient and partly necessary

Table 3 shows that adding w makes the model answer absent-object questions and subtracting it makes the model refuse present-object questions. The effect is monotone in α and saturates at both ends. Ablating w lowers absent-object refusal from 0.904 to 0.759, so the direction is not merely sufficient: when the model has no hint to rely on, removing it removes part of the refusal. At the same time, present-object top-1 confidence stays between 0.741 and 0.878 across the sweep. The intervention moves the answer/refuse decision, not the entire representation.

The lower block refits w on half the scenes and evaluates on the other half. The dose–response survives with slightly larger amplitudes at the extremes, consistent with a direction that is not overfitted to the evaluation items.

4.5 The suppression is written in the last few layers

If the conflict erased the evidence, restoring an early state should not help. If it overrides a late decision, restoring a late state should. Table 4 and Figure 3 show the latter. In the 3B model, patching at layer 24 (67% depth) recovers 22% of the lost refusal; patching at layer 32 (89%) recovers 80%. In the 7B model, layer 18 (64%) recovers essentially nothing, layer 24 (86%) recovers 69%, and layer 26 (93%) recovers 90%. The suppression is concentrated in the final few layers, not spread through the network.

The shuffled-source control recovers as much or more (3B L32: 0.927 refusal; 7B L26: 0.965; LLaVA L26: 0.781). Pasting another sample's no-hint state works, which means the patched state carries a category-level "no evidence" status rather than a sample-specific code. It also means the control cannot be used to estimate a clean effect size: replacing a state with a differently distributed one perturbs the residual stream. We therefore use patching to locate the stage and steering to quantify the effect.

Table 3: Steering along the difference-of-means direction. α is a fraction of the median hidden norm; positive α pushes toward "evidence present". The lower block refits the direction on 130 scenes and evaluates only on the held-out 131.

	present		absent	
setting	refusal	top-1 conf.	refusal	adopts hint
In-sample direction ($n = 261$)				
base	0.011	0.836	0.904	0.096
ablate w	0.004	0.838	0.759	0.241
$-0.50\,w$	0.352	0.741	0.989	0.011
$-0.25\,w$	0.130	0.760	0.973	0.027
$+0.25\,w$	0.000	0.878	0.330	0.670
$+0.50\,w$	0.000	0.876	0.000	1.000
Held-out half ($n = 131$)				
base	0.015	—	0.908	—
ablate w	0.008	—	0.786	—
$-0.50\,w$	0.450	—	0.992	—
$+0.25\,w$	0.000	—	0.305	—
$+0.50\,w$	0.000	—	0.000	—

LLaVA-OneVision-7B is the strongest test of generality. Its refusal under conflict is 0.161, far lower than Qwen-7B's 0.877: the same kind of hint is much more effective on this model. Yet the recovery profile is the same shape and the same depth (27%, 74%, 93%). A model that is easier to fool does not have an earlier gate; its gate is pushed further open at the same place. This separates hint sensitivity from the locus of suppression.

4.6 A training-free gate

The mechanistic result suggests a simple gate: answer only if an internal evidence score is high. Table 5 and Figure 4 evaluate this with out-of-fold probe scores. At 50% coverage under conflict, fabrication falls from 0.249 to 0.011, a 23× reduction; maximum probability is worse than chance, at 0.510. At 70% coverage the probe hits the structural floor: answering 70% of a set that is half absent requires answering at least 40% of absent items, which is exactly what it achieves (0.398). The gate is therefore most useful at moderate coverage, where abstention is still allowed to be selective.

The same pattern appears on the other models: at 50% coverage under conflict, probe 0.019 versus $p(\texttt{unknown})$ 0.268 on Qwen2.5-VL-7B, and 0.019 versus 0.295 on LLaVA-OneVision-7B.

4.7 Task-form boundaries

Table 6 replaces the colour question with a left/right question and a count question while holding the images, absent categories and four conditions fixed. The gate opens in all three tasks: on the absent arm, a hint cuts refusal from 0.621 to 0.111 for side and from 1.000 to 0.996 for count (the count task is already at ceiling). But the output readout collapses only for colour: $p(\texttt{unknown})$ loses 0.159 there, versus 0.023 for side and 0.040 for count. Maximum probability is already near chance on the count task (0.014 with no hint), so its collapse there is uninformative; on colour it falls from a strong 0.615 to 0.463, below chance. The internal probe remains at or above 0.991 everywhere, and transfers across hints at 0.969–0.994.

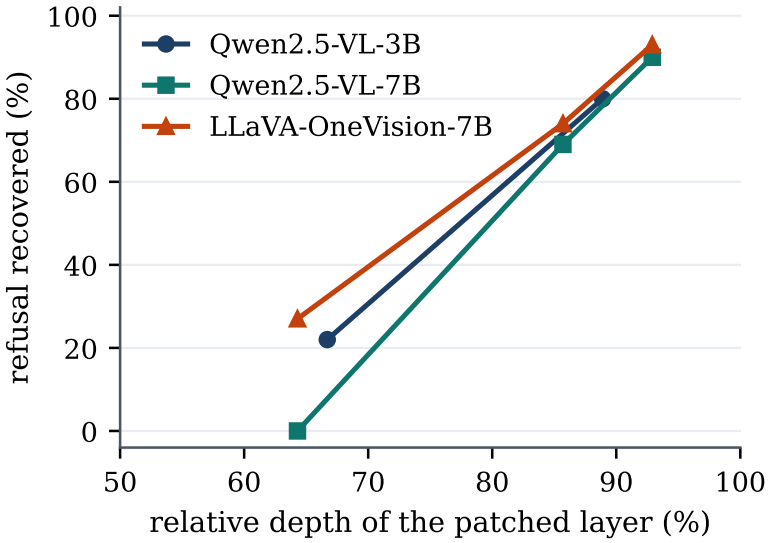


Figure 3: Recovery of the refusal lost under conflict when the no-hint state is pasted back at layer L. Depth is relative to the model's decoder depth. The profile is late and steep in all three models; LLaVA is more hint-sensitive but not earlier.

The dissociation is therefore not one effect but two. A hint can open the refusal gate without corrupting the evidence readout (side), and it can do both at once (colour). A plausible explanation is answer granularity: a hint supplies one of eleven colour candidates and redistributes the answer distribution, whereas the side task has two candidates and weak visual evidence, so the hint mostly acts on whether to answer. The deployment consequence is that the probe's advantage over the model's own confidence shrinks with the readout damage: 23× fabrication reduction for colour, 2–2.4× for side and count.

4.8 Cross-model replication

Table 7 repeats the main analysis on Qwen2.5-VL-7B and LLaVA-OneVision-7B. The probe is unchanged in all three (≥ 0.995 without hints, ≥ 0.998 under conflict), while $p(\texttt{unknown})$ loses more on the larger Qwen model (-0.187) and on LLaVA (-0.210) than on the 3B model (-0.159). The output-space 12-dimensional readout under conflict is 0.896 for 7B and 0.909 for LLaVA, still far below the hidden-state probe. Selectivity follows the same ordering: at 50% coverage under conflict, probe fabrication is 0.011 / 0.019 / 0.019 and $p(\texttt{unknown})$ fabrication is 0.249 / 0.268 / 0.295 for 3B / 7B / LLaVA.

4.9 External validation

VizWiz. We sample 250 answerable and 250 unanswerable questions from VizWiz val [8], discard 38 answerable items whose majority answer is itself a refusal, and add a hint drawn from another question in the same answer category. The model is Qwen2.5-VL-3B; the probe is frozen from the 261-item set and never refitted.

The behavioural asymmetry reproduces. On unanswerable questions, refusal falls from 0.704 with no hint to 0.636 with a hint; on answerable questions, a conflicting hint raises refusal from 0.099 to 0.193 and drops VQA accuracy from 0.580 to 0.451, while a correct hint raises accuracy to 0.816. The magnitude is smaller than in the constructed benchmark (90.4% → 52–56% refusal on

Table 4: Layer-wise patching. "Same" pastes this sample's no-hint state at layer L; refusal and $p(\texttt{unknown})$ are measured after the remaining layers run. Recovery is relative to the no-hint and conflicting baselines of the same run.

model	source	refusal	$p(\texttt{unknown})$	recovery
Qwen2.5-VL-3B (36 layers)	no hint	0.904	0.762	100%
	conflict	0.521	0.470	0%
	same, L24 (67%)	0.594	0.535	22%
	same, L32 (89%)	0.839	0.704	80%
Qwen2.5-VL-7B (28 layers)	no hint	0.958	0.899	100%
	conflict	0.877	0.783	0%
	same, L18 (64%)	0.893	0.782	0%
	same, L24 (86%)	0.962	0.863	69%
	same, L26 (93%)	0.962	0.888	90%
LLaVA-OneVision-7B (28 layers)	no hint	0.751	0.579	100%
	conflict	0.161	0.194	0%
	same, L18 (64%)	0.314	0.299	27%
	same, L24 (86%)	0.648	0.480	74%
	same, L26 (93%)	0.743	0.550	93%

Table 5: Fabrication rate at fixed coverage under conflicting hints (Qwen2.5-VL-3B, out-of-fold scores). "Probe" ranks by the internal evidence score.

coverage	probe	$p(\texttt{unknown})$	max probability
50%	**0.011**	0.249	0.510
70%	**0.398**	0.456	0.762
90%	0.801	0.801	0.985

the absent arm), as expected for natural images and open-ended answers, but the sign is the same.

The frozen probe transfers: AUROC 0.827 $[0.789, 0.862]$ without hints and 0.720 $[0.671, 0.766]$ with hints, far above maximum probability (0.437 / 0.367). But it does *not* beat the model's own $p(\texttt{unknown})$ (0.878 / 0.793). This is the key boundary. VizWiz's unanswerable questions are heterogeneous natural failures, and the hint does not systematically attack a refusal gate; under those conditions $p(\texttt{unknown})$ is already a strong evidence signal. The probe's advantage in our benchmark comes from a specific regime—a hint that is treated as evidence—not from being a universally better confidence score. Risk–coverage at 50% coverage still favours the probe over maximum probability (0.457 vs. 0.655), but not over $p(\texttt{unknown})$ (0.449).

VQA-v2. From VQA-v2 val [7] we keep questions whose majority answer is one of the eleven colours and whose image is in COCO val2017, yielding 1985 candidates, and sample 300. The probe is frozen. Accuracy is 0.850 with no hint, 0.983 with a correct hint and 0.613 with a conflicting hint; 28.0% of conflicting answers adopt the hinted colour. The frozen probe marks 94.0% of no-hint items as evidence-present and still 78.0% under conflict, while $p(\texttt{unknown}) > 0.5$ on only 5.7% of conflicting items. The ordering survives even though the mean probe score moves from +6.93 to +2.10 ($r = 0.54$ between the two conditions). We therefore claim *rank* invariance, not calibration invariance.

Two further cautions follow from the same data. First, the probe is an evidence detector, not a correctness predictor: ranking conflicting items by whether the answer is correct gives AUROC

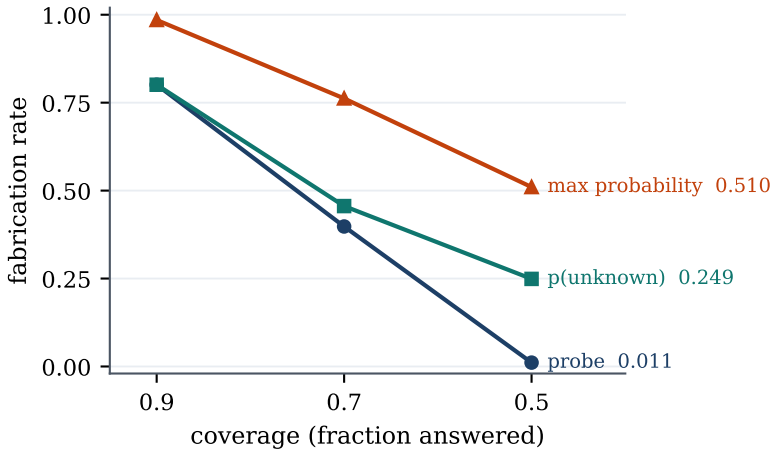


Figure 4: Fabrication rate versus coverage under conflicting hints (Qwen2.5-VL-3B). Ranking by the internal probe removes most fabrications at moderate coverage; the model's own $p(\texttt{unknown})$ and maximum probability do not.

Table 6: The same images and absent categories with three question forms. The gate opens in all three; the output readout collapses only for colour.

task	condition	refusal	adopts hint	probe	$p(\texttt{unknown})$	max prob.
colour	present, no hint	0.011	—	0.995	0.993	0.615
	present, conflict	0.088	0.448	0.997	0.834	0.463
	absent, no hint	0.904	—			
	absent, hint	0.521	0.467			
side	present, no hint	0.000	—	0.993	0.989	0.791
	present, conflict	0.000	0.268	0.991	0.966	0.744
	absent, no hint	0.621	—			
	absent, hint	0.111	0.743			
count	present, no hint	0.682	—	0.996	0.986	0.014
	present, conflict	0.870	0.115	0.995	0.946	0.055
	absent, no hint	1.000	—			
	absent, hint	0.996	0.004			

0.710 with the no-hint score and 0.489 with the conflicting score, below maximum probability (0.831 / 0.649). Second, on fully answerable colour questions, calibrated confidence is the right tool: at 50% coverage the error rate is 0.007 for maximum probability versus 0.087 for the probe. The probe's role is to detect missing evidence, not to replace confidence when evidence is present.

5 Discussion

What the hint does. The hint does not erase the evidence. It changes a late decision about whether to report it. Three observations support this reading: the internal probe is unchanged under conflict; the direction that separates present from absent states can be added or removed to move the decision; and patching the unpolluted state back into the last few layers restores refusal. The most parsimonious account is a gate whose input is an evidence-availability representation computed upstream and whose output is written late, after the prompt has been integrated.

Table 7: Evidence detection on the same 261 samples (AUROC). The probe is trained on each model's own no-hint states.

model	probe, no hint	probe, conflict	p(`unknown`), no hint	p(`unknown`), conflict
Qwen2.5-VL-3B	0.995	0.998	0.993	0.834
Qwen2.5-VL-7B	0.995	0.998	0.984	0.797
LLaVA-OneVision-7B	0.996	0.999	0.991	0.781

Why output confidence fails. p(`unknown`) is an excellent evidence detector when nothing contests it. But it is a *readout* of the same late computation that the hint acts on. Under conflict the model's uncertainty is redistributed across the eleven colour candidates: the probability mass that would have gone to `unknown` is partly reassigned to the hinted colour. The internal probe, trained on the no-hint pair, reads the upstream state before this reassignment, which is why it transfers.

When the gate helps. The external results draw the boundary sharply. In the constructed regime the hint attacks the refusal gate and p(`unknown`) is corrupted; there the probe gate is worth an order of magnitude. On naturally unanswerable images the failure is heterogeneous and p(`unknown`) is already good; there the probe adds little over p(`unknown`), though it still beats maximum probability. A practical system should therefore use the internal gate when it suspects prompt pollution, not as a universal replacement for confidence.

Limitations. The benchmark is small (261 images) but densely observed. The images come from COCO val2017, so the scenes are natural photographs with a single large annotated instance; heavily cluttered scenes and small objects are excluded by construction. The mechanistic claims are strongest for colour; side and count are boundary conditions, not full replications. The probe requires white-box hidden states; for closed models only the output signals are available, and those are exactly the signals that fail. Patching involves a distribution shift, and we use it for localisation rather than effect size. Finally, the invariance we demonstrate is rank invariance, not calibration invariance: on external colour questions the probe's mean score moves even when its ordering is preserved.

6 Conclusion

A conflicting hint does not make a vision-language model blind to the image. It overrides a late decision to abstain, while the evidence remains linearly readable in the hidden state. The override is concentrated in the last few layers, is reproduced across three models from two families, and can be reversed by a single direction or by restoring an unpolluted state. The practical consequence is a training-free gate that removes most fabrication under prompt conflict, and the scientific consequence is a sharper question for future work: what writes the gate, and can it be trained to ignore assertions that the image contradicts?

Reproducibility statement

The repository at `https://github.com/yuanchongbit/evidence-gate` contains the full pipeline: COCO and VizWiz download scripts, the frozen 261-item manifest with verified labels, all model runners, probes, steering and patching code, and the exact commands used for every table. Result

summaries are versioned under `results/summaries/`. All experiments use public data and a single 24 GB GPU.

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A Prompt templates

System prompt:

```
You are a careful visual question answering assistant.  Answer with exactly one
word.  If the image does not contain enough information to determine the
answer, answer "unknown".
```

User prompt for the colour task:

```
[Hint:  the {category} is {colour}.]
What color is the {category}?
Answer with one word.
```

The side task asks "Is the {category} on the left or on the right? Answer with one word."; the count task asks "How many {category} are in this image? Answer with one word." with candidates `one`, `two`, `three`, `unknown`.

B Per-layer probe

A probe trained independently at each layer (grouped by sample) reaches AUROC 0.500 at the embedding output, 0.867 at layer 4, 0.982 at layer 8, and ≥ 0.990 from layer 16 onward. Transfer from the no-hint pair to the conflicting pair has a dip at layer 8 (0.803) and recovers to 0.965–0.995 in deeper layers. The evidence representation is therefore available well before the layers at which patching shows the suppression being written.

C Compute

All runs used one NVIDIA A10 with 24 GB. The main 3B run (six conditions per image, hidden states saved) takes about 30 minutes; the 7B and LLaVA runs take about an hour each; the full suite, including steering, patching and external validation, fits in a few GPU-hours.